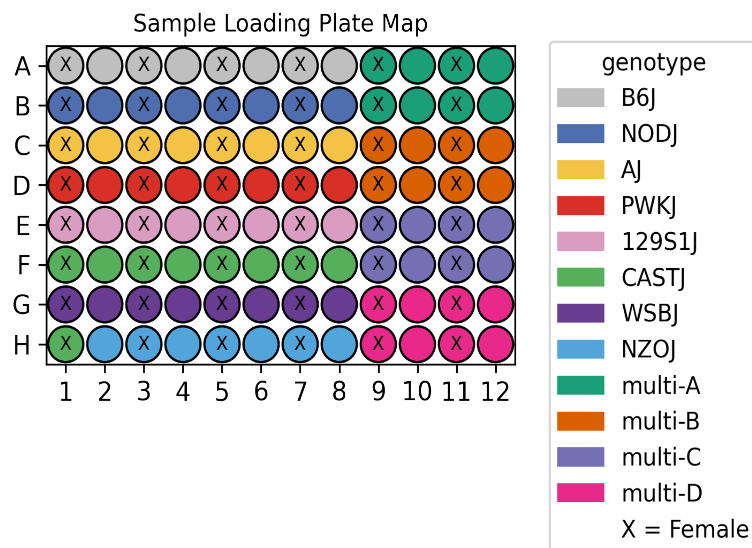
UCI/Caltech IGVF Dataset Pipeline Report

	igvf_003
Kit	WT_mega
Chemistry	Parse v2
Sample type	nuclei
Genotypes	B6J NODJ AJ PWKJ 129S1J CASTJ WSBJ NZOJ
Tissues	CortexHippocampus Heart
Subpool	Subpool_9
Measurment Set	IGVFDS1440FXHP
Exome capture subpool	No
Expected nuclei	67,000



1. Sample multiplexing

CortexHippocampus was the main tissue on the plate while Heart was multiplexed.

Table 1: Genotype multiplexing key

Pair	Multiplexed Genotype 1	Multiplexed Genotype 2
multi-A	B6J	NODJ
multi-B	AJ	PWKJ
multi-C	129S1J	CASTJ
multi-D	NZOJ	WSBJ

2. Alignment

A total of 1,185,598,644 reads were aligned using GENCODE vM32 with mm39 assembly.

Table 2: Pseudoalignment stats

Subpool	Total reads	Total UMIs	Fraction duplicated UMIs	Total pseudoaligned	Pct. pseudoaligned
Subpool_9	1,185,598,644	538,929,428	0.377	865,022,475	73.0

3. Cell recovery

A total of 110,877 nuclei in Subpool_9 have >=200 UMIs and 56,766 nuclei have >= 500 UMIs.

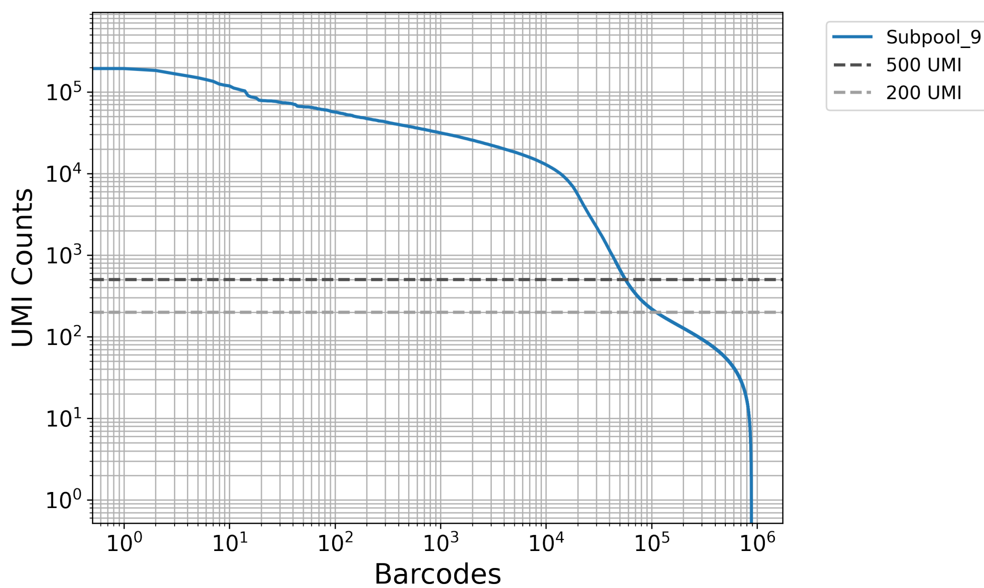


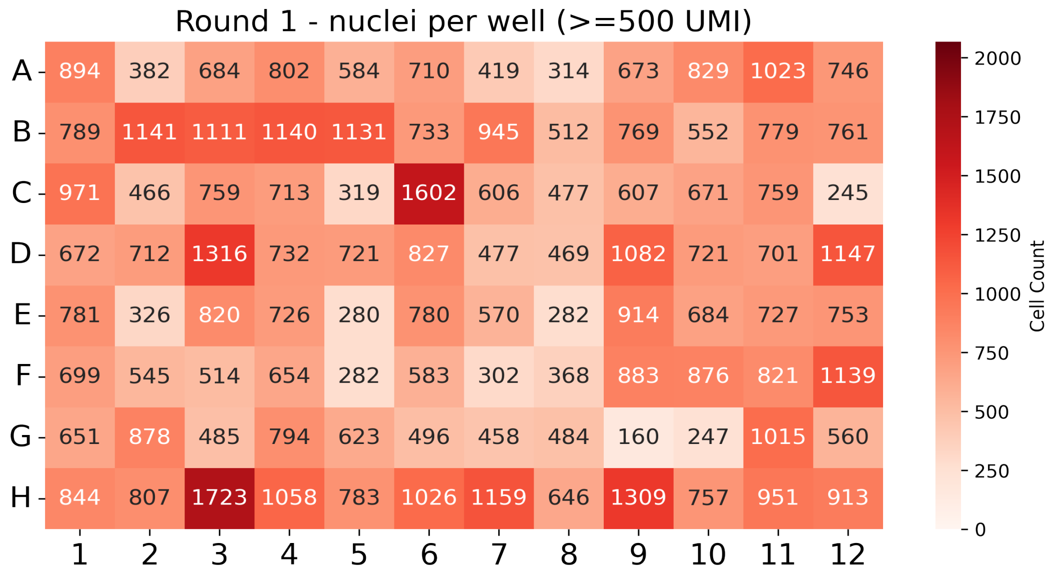
Table 3: QC stats for nuclei >=200 UMI

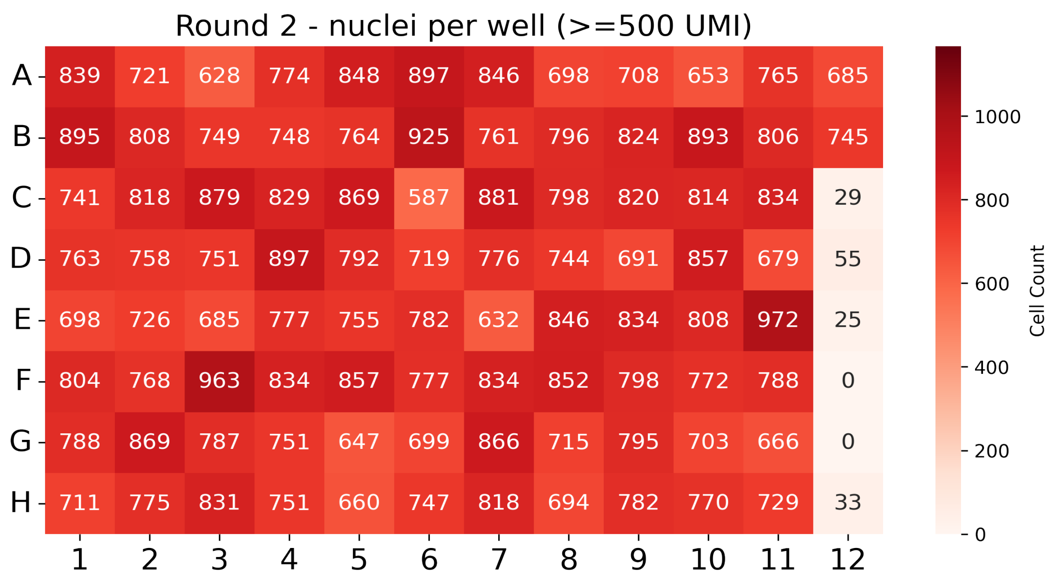
Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_9	110,877	526	3,444.4	362	1,092.8

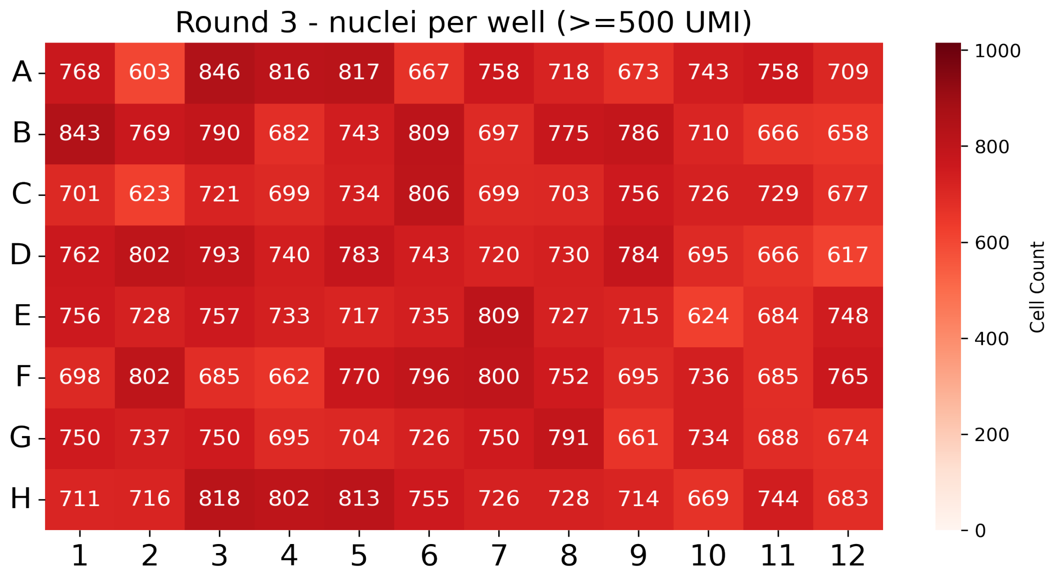
Table 4: QC stats for nuclei >=500 UMI

Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_9	56,766	2,507	6,451.4	1,314	1,924.3

In this combinatorial barcoding assay, nuclei are barcoded in 96-well plates over three rounds. The first round assigns sample-specific barcodes, while the next two are applied to the pooled mix. Even distribution in the first round is important to ensure consistent sample representation.


Round 1 - nuclei per well (>=500 UMI)
A 894 382 684 802 584 710 419 314 673 829 1023 746
B 789 1141 1111 1140 1131 733 945 512 769 552 779 761
C 971 466 759 713 319 1602 606 477 607 671 759 245
D 672 712 1316 732 721 827 477 469 1082 721 701 1147
E 781 326 820 726 280 780 570 282 914 684 727 753
F 699 545 514 654 282 583 302 368 883 876 821 1139
G 651 878 485 794 623 496 458 484 160 247 1015 560
H 844 807 1723 1058 783 1026 1159 646 1309 757 951 913
1 2 3 4 5 6 7 8 9 10 11 12
2000
1750
1500
1250
1000
750
500
250
0
Cell Count


Round 2 - nuclei per well (>=500 UMI)
A 839 721 628 774 848 897 846 698 708 653 765 685
B 895 808 749 748 764 925 761 796 824 893 806 745
C 741 818 879 829 869 587 881 798 820 814 834 29
D 763 758 751 897 792 719 776 744 691 857 679 55
E 698 726 685 777 755 782 632 846 834 808 972 25
F 804 768 963 834 857 777 834 852 798 772 788 0
G 788 869 787 751 647 699 866 715 795 703 666 0
H 711 775 831 751 660 747 818 694 782 770 729 33
1 2 3 4 5 6 7 8 9 10 11 12
1000
800
600
400
200
0
Cell Count



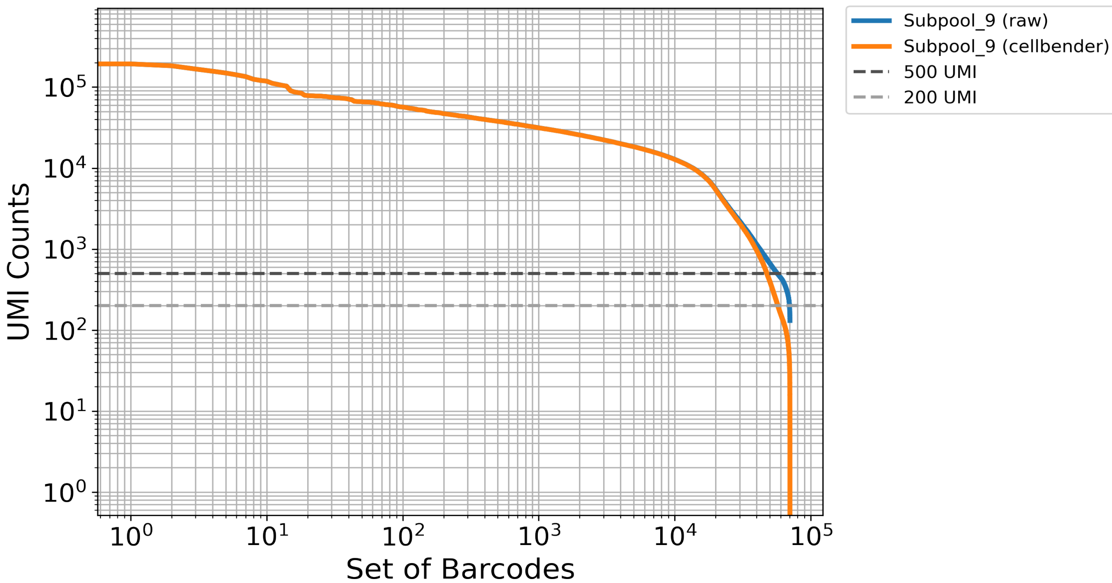
4. Background removal

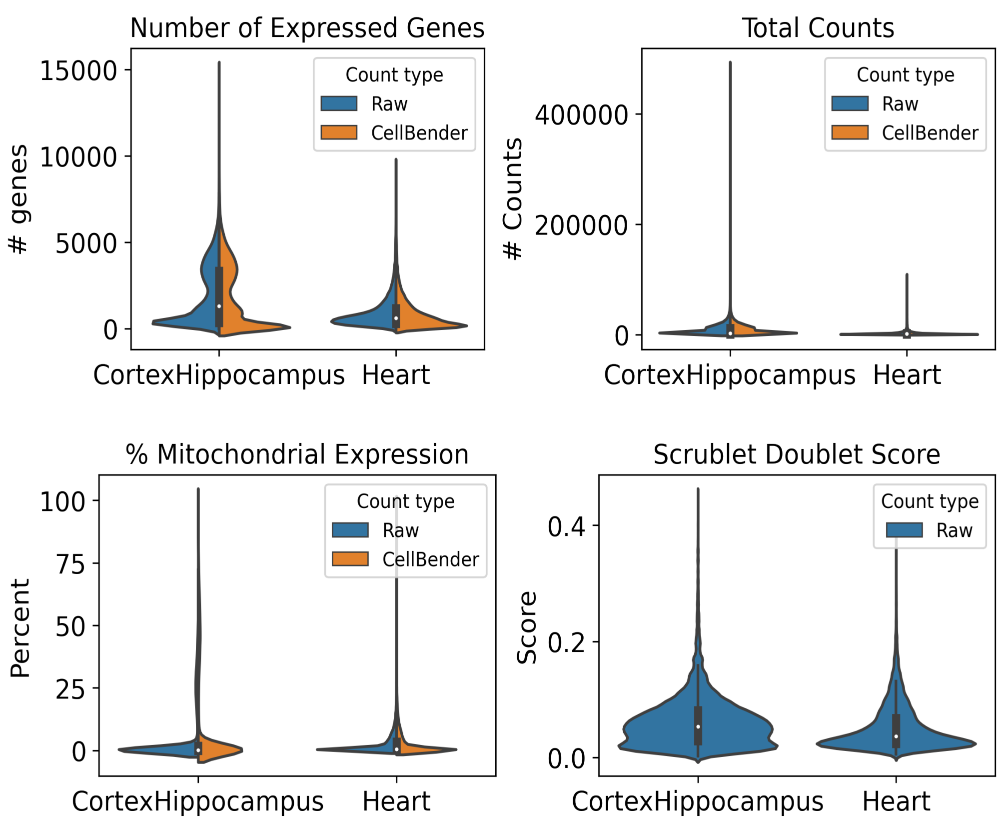
Table 5: Cellbender parameters

Subpool	droplets_included	learning_rate	expected_cells
Subpool_9	200000	5e-05	NA

Table 6: Cellbender stats

Subpool	Percent counts removed	Found nuclei	Mean denoised counts
Subpool_9	3.2	70,331	5,110.4





5. Barcode sample map

The first round corresponds to sample barcoding. In Parse Bio assays, two barcodes are included in the first round: a random hexamer (R) and oligo-dT (T) barcode per well (parse barcode type). This reduces 3-prime bias and allows for the capture of full-length transcripts. These sample level barcodes can be found in the barcode-1 region in the seqspec, and begin at position 16 (0-based) in the full 24-nt cell barcode. For the WT_mega kit, samples are distributed across 96 wells in round 1. The following table includes our lab sample ID as sample descriptions and IGVF sample accessions.

Table 7: Barcode sample map

barcode	sample accession	sample description	position	parse barcode type	well
CATCATCC	IGVFSM2747ZOJJ,IGVFSM9909OKAR	016_B6J_10F_03	16	R	A1
CATTCCTA	IGVFSM2747ZOJJ,IGVFSM9909OKAR	016_B6J_10F_03	16	T	A1
CTGCTTTG	IGVFSM5751LMFR,IGVFSM9616JAGA	017_B6J_10M_03	16	R	A2
CTTCATCA	IGVFSM5751LMFR,IGVFSM9616JAGA	017_B6J_10M_03	16	T	A2
CTAAGGGA	IGVFSM4584AIMA,IGVFSM6999WOUL	018_B6J_10F_03	16	R	A3
CCTATATC	IGVFSM4584AIMA,IGVFSM6999WOUL	018_B6J_10F_03	16	T	A3
GCTTATAG	IGVFSM3502KEWR,IGVFSM9565QVTN	019_B6J_10M_03	16	R	A4
ACATTTAC	IGVFSM3502KEWR,IGVFSM9565QVTN	019_B6J_10M_03	16	T	A4
TCTGATCC	IGVFSM6578SAUC,IGVFSM9764WOWB	020_B6J_10F_03	16	R	A5
ACTTAGCT	IGVFSM6578SAUC,IGVFSM9764WOWB	020_B6J_10F_03	16	T	A5
TCTCTTGG	IGVFSM2795QIAK,IGVFSM3360ORUF	021_B6J_10M_03	16	R	A6
CCAATTCT	IGVFSM2795QIAK,IGVFSM3360ORUF	021_B6J_10M_03	16	T	A6
CAATTTCC	IGVFSM0587AFGX,IGVFSM7596ZIPA	024_B6J_10F_03	16	R	A7
GCCTATCT	IGVFSM0587AFGX,IGVFSM7596ZIPA	024_B6J_10F_03	16	T	A7
AGTCTCTT	IGVFSM1457MNSR,IGVFSM4108RLII	025_B6J_10M_03	16	R	A8
ATGCTGCT	IGVFSM1457MNSR,IGVFSM4108RLII	025_B6J_10M_03	16	T	A8
TGCTGCTC	IGVFSM2397OXJE	016_B6J_10F_06	16	R	A9
CATTTACA	IGVFSM2397OXJE	016_B6J_10F_06	16	T	A9
TGCTGCTC	IGVFSM0400KSXT	066_NODJ_10F_06	16	R	A9
CATTTACA	IGVFSM0400KSXT	066_NODJ_10F_06	16	T	A9
GTATTTCC	IGVFSM6821QIOI	017_B6J_10M_06	16	R	A10
ACTCGTAA	IGVFSM6821QIOI	017_B6J_10M_06	16	T	A10
GTATTTCC	IGVFSM8486IRSE	067_NODJ_10M_06	16	R	A10
ACTCGTAA	IGVFSM8486IRSE	067_NODJ_10M_06	16	T	A10
TTCCTGTG	IGVFSM9283STYP	018_B6J_10F_06	16	R	A11
CCTTTGCA	IGVFSM9283STYP	018_B6J_10F_06	16	T	A11
TTCCTGTG	IGVFSM6061AIFT	068_NODJ_10F_06	16	R	A11

CCTTTGCA	IGVFSM6061AIFT	068_NODJ_10F_06	16	T	A11
GCTGCTTC	IGVFSM3819GVRA	019_B6J_10M_06	16	R	A12
ACTCCTGC	IGVFSM3819GVRA	019_B6J_10M_06	16	T	A12
GCTGCTTC	IGVFSM9555SHCP	069_NODJ_10M_06	16	R	A12
ACTCCTGC	IGVFSM9555SHCP	069_NODJ_10M_06	16	T	A12
TATGTGTC	IGVFSM1810KYOB,IGVFSM5439QLWY	066_NODJ_10F_03	16	R	B1
ATTTGGCA	IGVFSM1810KYOB,IGVFSM5439QLWY	066_NODJ_10F_03	16	T	B1
CAATTCTC	IGVFSM3477RPDP,IGVFSM6514BRQT	067_NODJ_10M_03	16	R	B2
TTATTCTG	IGVFSM3477RPDP,IGVFSM6514BRQT	067_NODJ_10M_03	16	T	B2
TGGTCTCC	IGVFSM0483CXTW,IGVFSM2315DNPR	068_NODJ_10F_03	16	R	B3
TCATGCTC	IGVFSM0483CXTW,IGVFSM2315DNPR	068_NODJ_10F_03	16	T	B3
GCTCTTTA	IGVFSM2210CUOC,IGVFSM3655BTER	069_NODJ_10M_03	16	R	B4
CATACTTC	IGVFSM2210CUOC,IGVFSM3655BTER	069_NODJ_10M_03	16	T	B4
GCTGCATG	IGVFSM1916FXAR,IGVFSM4132TDOU	070_NODJ_10F_03	16	R	B5
CCGTTCTA	IGVFSM1916FXAR,IGVFSM4132TDOU	070_NODJ_10F_03	16	T	B5
ACTCATTT	IGVFSM2219SJUB,IGVFSM9423CHGE	071_NODJ_10M_03	16	R	B6
GCTTCATA	IGVFSM2219SJUB,IGVFSM9423CHGE	071_NODJ_10M_03	16	T	B6
AGTCTTGG	IGVFSM0749UUSU,IGVFSM1724CQMW	072_NODJ_10F_03	16	R	B7
CTCTGTGC	IGVFSM0749UUSU,IGVFSM1724CQMW	072_NODJ_10F_03	16	T	B7
GGTTCTTC	IGVFSM5506IVPG,IGVFSM9890CMQT	073_NODJ_10M_03	16	R	B8
CCCTTATA	IGVFSM5506IVPG,IGVFSM9890CMQT	073_NODJ_10M_03	16	T	B8
TCATGTTG	IGVFSM9188TGLO	020_B6J_10F_06	16	R	B9
ACTGCTCT	IGVFSM9188TGLO	020_B6J_10F_06	16	T	B9
TCATGTTG	IGVFSM2548HETT	070_NODJ_10F_06	16	R	B9
ACTGCTCT	IGVFSM2548HETT	070_NODJ_10F_06	16	T	B9
ATTTTGCC	IGVFSM4152MQAV	021_B6J_10M_06	16	R	B10
CTCTAATC	IGVFSM4152MQAV	021_B6J_10M_06	16	T	B10
ATTTTGCC	IGVFSM6976MFYC	071_NODJ_10M_06	16	R	B10
CTCTAATC	IGVFSM6976MFYC	071_NODJ_10M_06	16	T	B10
CTTCTGTA	IGVFSM1715AZGK	024_B6J_10F_06	16	R	B11
ACCCTTGC	IGVFSM1715AZGK	024_B6J_10F_06	16	T	B11
CTTCTGTA	IGVFSM1795GGVQ	072_NODJ_10F_06	16	R	B11
ACCCTTGC	IGVFSM1795GGVQ	072_NODJ_10F_06	16	T	B11
GTCCATCT	IGVFSM0659JNLJ	025_B6J_10M_06	16	R	B12
ATCTTAGG	IGVFSM0659JNLJ	025_B6J_10M_06	16	T	B12

GTCCATCT	IGVFSM9138HBOW	073_NODJ_10M_06	16	R	B12
ATCTTAGG	IGVFSM9138HBOW	073_NODJ_10M_06	16	T	B12
GCTATCTC	IGVFSM2423BNZZ,IGVFSM4370URZQ	026_AJ_10F_03	16	R	C1
CATGTCTC	IGVFSM2423BNZZ,IGVFSM4370URZQ	026_AJ_10F_03	16	T	C1
TAGTTTCC	IGVFSM2957XPNR,IGVFSM7221OIET	029_AJ_10M_03	16	R	C2
TCATTGCA	IGVFSM2957XPNR,IGVFSM7221OIET	029_AJ_10M_03	16	T	C2
TCCATTAT	IGVFSM7981HLTV,IGVFSM8529NIEC	028_AJ_10F_03	16	R	C3
ACACCTTT	IGVFSM7981HLTV,IGVFSM8529NIEC	028_AJ_10F_03	16	T	C3
AGGATTAA	IGVFSM6144DRYM,IGVFSM8120ZQGC	031_AJ_10M_03	16	R	C4
AATTTCTC	IGVFSM6144DRYM,IGVFSM8120ZQGC	031_AJ_10M_03	16	T	C4
AATCTTTC	IGVFSM0425PEQP,IGVFSM9442VMKA	030_AJ_10F_03	16	R	C5
ATTCATGG	IGVFSM0425PEQP,IGVFSM9442VMKA	030_AJ_10F_03	16	T	C5
GTCATATG	IGVFSM0406DUWH,IGVFSM9867TADD	033_AJ_10M_03	16	R	C6
ACTTTACC	IGVFSM0406DUWH,IGVFSM9867TADD	033_AJ_10M_03	16	T	C6
GTGCTTCC	IGVFSM3195CJCF,IGVFSM6726YGIG	032_AJ_10F_03	16	R	C7
CTTCTAAC	IGVFSM3195CJCF,IGVFSM6726YGIG	032_AJ_10F_03	16	T	C7
ATGTGTTG	IGVFSM4131CTSJ,IGVFSM4651QBMQ	035_AJ_10M_03	16	R	C8
CTATTTCA	IGVFSM4131CTSJ,IGVFSM4651QBMQ	035_AJ_10M_03	16	T	C8
CCATCTTG	IGVFSM7523CIFZ	026_AJ_10F_06	16	R	C9
TCTCATGC	IGVFSM7523CIFZ	026_AJ_10F_06	16	T	C9
CCATCTTG	IGVFSM2489HRKN	076_PWKJ_10F_06	16	R	C9
TCTCATGC	IGVFSM2489HRKN	076_PWKJ_10F_06	16	T	C9
TACTGTCT	IGVFSM5294ZSIL	029_AJ_10M_06	16	R	C10
ATCCTTAC	IGVFSM5294ZSIL	029_AJ_10M_06	16	T	C10
TACTGTCT	IGVFSM9967KFUW	077_PWKJ_10M_06	16	R	C10
ATCCTTAC	IGVFSM9967KFUW	077_PWKJ_10M_06	16	T	C10
TTCATCGC	IGVFSM1802ZPXC	028_AJ_10F_06	16	R	C11
TAAATATC	IGVFSM1802ZPXC	028_AJ_10F_06	16	T	C11
TTCATCGC	IGVFSM9748KTTY	078_PWKJ_10F_06	16	R	C11
TAAATATC	IGVFSM9748KTTY	078_PWKJ_10F_06	16	T	C11
ACTGTGGG	IGVFSM4608HMDV	031_AJ_10M_06	16	R	C12
TTACCTGC	IGVFSM4608HMDV	031_AJ_10M_06	16	T	C12
ACTGTGGG	IGVFSM6943ARXK	079_PWKJ_10M_06	16	R	C12
TTACCTGC	IGVFSM6943ARXK	079_PWKJ_10M_06	16	T	C12
TCTGTGCC	IGVFSM0918MYKM,IGVFSM7550NDQP	076_PWKJ_10F_03	16	R	D1

CACTTTCA	IGVFSM0918MYKM,IGVFSM7550NDQP	076_PWKJ_10F_03	16	T	D1
TCAATCTC	IGVFSM0563DEOO,IGVFSM6013VRIK	077_PWKJ_10M_03	16	R	D2
CACCTTTA	IGVFSM0563DEOO,IGVFSM6013VRIK	077_PWKJ_10M_03	16	T	D2
GTCCTCTG	IGVFSM0623LMKH,IGVFSM6356IYPX	078_PWKJ_10F_03	16	R	D3
CTGACTTC	IGVFSM0623LMKH,IGVFSM6356IYPX	078_PWKJ_10F_03	16	T	D3
TTACATTC	IGVFSM2048SMTE,IGVFSM4224SVKL	079_PWKJ_10M_03	16	R	D4
CATTTGGA	IGVFSM2048SMTE,IGVFSM4224SVKL	079_PWKJ_10M_03	16	T	D4
ATTCTGTC	IGVFSM1463UXNU,IGVFSM3905UDST	080_PWKJ_10F_03	16	R	D5
GCTCTACT	IGVFSM1463UXNU,IGVFSM3905UDST	080_PWKJ_10F_03	16	T	D5
TGTGTATG	IGVFSM0454IWHA,IGVFSM1764RZHQ	081_PWKJ_10M_03	16	R	D6
GTTACGTA	IGVFSM0454IWHA,IGVFSM1764RZHQ	081_PWKJ_10M_03	16	T	D6
TCCATTTG	IGVFSM2111IGAF,IGVFSM7111UROK	084_PWKJ_10F_03	16	R	D7
CCTGTTGC	IGVFSM2111IGAF,IGVFSM7111UROK	084_PWKJ_10F_03	16	T	D7
TTAGCTTC	IGVFSM1227KXDL,IGVFSM5036ECAW	085_PWKJ_10M_03	16	R	D8
CTATCATC	IGVFSM1227KXDL,IGVFSM5036ECAW	085_PWKJ_10M_03	16	T	D8
GTGCTTGA	IGVFSM5549SWHL	030_AJ_10F_06	16	R	D9
GCTATCAT	IGVFSM5549SWHL	030_AJ_10F_06	16	T	D9
GTGCTTGA	IGVFSM8513LDPM	080_PWKJ_10F_06	16	R	D9
GCTATCAT	IGVFSM8513LDPM	080_PWKJ_10F_06	16	T	D9
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ACATTCAT	IGVFSM7128WVEJ	033_AJ_10M_06	16	T	D10
GTTTGTGA	IGVFSM3449RSOM	081_PWKJ_10M_06	16	R	D10
ACATTCAT	IGVFSM3449RSOM	081_PWKJ_10M_06	16	T	D10
GAAATTAG	IGVFSM2048BQHM	032_AJ_10F_06	16	R	D11
TTCGCTAC	IGVFSM2048BQHM	032_AJ_10F_06	16	T	D11
GAAATTAG	IGVFSM5687CRRX	084_PWKJ_10F_06	16	R	D11
TTCGCTAC	IGVFSM5687CRRX	084_PWKJ_10F_06	16	T	D11
GCAAATTC	IGVFSM2812EMHZ	035_AJ_10M_06	16	R	D12
CATTCTAC	IGVFSM2812EMHZ	035_AJ_10M_06	16	T	D12
GCAAATTC	IGVFSM2657UOQA	085_PWKJ_10M_06	16	R	D12
CATTCTAC	IGVFSM2657UOQA	085_PWKJ_10M_06	16	T	D12
GAGGTTGA	IGVFSM8124ZHLJ,IGVFSM9227EIZD	036_129S1J_10F_03	16	R	E1
CACTTATC	IGVFSM8124ZHLJ,IGVFSM9227EIZD	036_129S1J_10F_03	16	T	E1
CCTGTCTG	IGVFSM0860OBUU,IGVFSM2419NKNV	037_129S1J_10M_03	16	R	E2
ATAAGCTC	IGVFSM0860OBUU,IGVFSM2419NKNV	037_129S1J_10M_03	16	T	E2

GTGGGTTC	IGVFSM0225XPQG,IGVFSM1783ISPH	038_129S1J_10F_03	16	R	E3
TCATCCTG	IGVFSM0225XPQG,IGVFSM1783ISPH	038_129S1J_10F_03	16	T	E3
TTTGCATC	IGVFSM4535GZXJ,IGVFSM9603NVUP	041_129S1J_10M_03	16	R	E4
CCTGGTAT	IGVFSM4535GZXJ,IGVFSM9603NVUP	041_129S1J_10M_03	16	T	E4
AGGTAATA	IGVFSM1865RBAP,IGVFSM3592ZCYS	040_129S1J_10F_03	16	R	E5
TGGTATAC	IGVFSM1865RBAP,IGVFSM3592ZCYS	040_129S1J_10F_03	16	T	E5
GTGCCTTC	IGVFSM2592YCXF,IGVFSM4046THJJ	043_129S1J_10M_03	16	R	E6
TTGGGAGA	IGVFSM2592YCXF,IGVFSM4046THJJ	043_129S1J_10M_03	16	T	E6
ATGTTTCC	IGVFSM0923PDGB,IGVFSM8263PWJC	044_129S1J_10F_03	16	R	E7
ACTTCATC	IGVFSM0923PDGB,IGVFSM8263PWJC	044_129S1J_10F_03	16	T	E7
CTTAATTC	IGVFSM0221ZKMO,IGVFSM9934DIHO	045_129S1J_10M_03	16	R	E8
TCTCTAGC	IGVFSM0221ZKMO,IGVFSM9934DIHO	045_129S1J_10M_03	16	T	E8
TCTGGCTC	IGVFSM9118PKWD	036_129S1J_10F_06	16	R	E9
ATGCCCTT	IGVFSM9118PKWD	036_129S1J_10F_06	16	T	E9
TCTGGCTC	IGVFSM5819SOLY	086_CASTJ_10F_06	16	R	E9
ATGCCCTT	IGVFSM5819SOLY	086_CASTJ_10F_06	16	T	E9
CATCATTT	IGVFSM8184LOKP	037_129S1J_10M_06	16	R	E10
CCCAATTT	IGVFSM8184LOKP	037_129S1J_10M_06	16	T	E10
CATCATTT	IGVFSM0487BHBU	087_CASTJ_10M_06	16	R	E10
CCCAATTT	IGVFSM0487BHBU	087_CASTJ_10M_06	16	T	E10
GTTGTCTC	IGVFSM3629MXTV	038_129S1J_10F_06	16	R	E11
ACTATATA	IGVFSM3629MXTV	038_129S1J_10F_06	16	T	E11
GTTGTCTC	IGVFSM5847JXKA	088_CASTJ_10F_06	16	R	E11
ACTATATA	IGVFSM5847JXKA	088_CASTJ_10F_06	16	T	E11
ATCTTCTG	IGVFSM9345WJHN	041_129S1J_10M_06	16	R	E12
CTCTATAC	IGVFSM9345WJHN	041_129S1J_10M_06	16	T	E12
ATCTTCTG	IGVFSM4021BOPQ	089_CASTJ_10M_06	16	R	E12
CTCTATAC	IGVFSM4021BOPQ	089_CASTJ_10M_06	16	T	E12
TGTTTGCC	IGVFSM7195PKWY,IGVFSM7666QOZY	086_CASTJ_10F_03	16	R	F1
CTGTCTCA	IGVFSM7195PKWY,IGVFSM7666QOZY	086_CASTJ_10F_03	16	T	F1
TTCTGTCA	IGVFSM7317YYMK,IGVFSM8253LIZU	087_CASTJ_10M_03	16	R	F2
GACCTTTC	IGVFSM7317YYMK,IGVFSM8253LIZU	087_CASTJ_10M_03	16	T	F2
ACGGACTC	IGVFSM3708BQMH,IGVFSM8386KIXU	088_CASTJ_10F_03	16	R	F3
GATTTGGC	IGVFSM3708BQMH,IGVFSM8386KIXU	088_CASTJ_10F_03	16	T	F3
TTTGGTCA	IGVFSM6275QJJU,IGVFSM7470VZNE	089_CASTJ_10M_03	16	R	F4

CGTCTAGG	IGVFSM6275QJJU,IGVFSM7470VZNE	089_CASTJ_10M_03	16	T	F4
TATCCGGG	IGVFSM3391TANO,IGVFSM9194WULA	090_CASTJ_10F_03	16	R	F5
TACTCGAA	IGVFSM3391TANO,IGVFSM9194WULA	090_CASTJ_10F_03	16	T	F5
TGTCATTC	IGVFSM3523HTHO,IGVFSM4618XBBU	091_CASTJ_10M_03	16	R	F6
CAGCCTTT	IGVFSM3523HTHO,IGVFSM4618XBBU	091_CASTJ_10M_03	16	T	F6
ATTCTCTG	IGVFSM7747QNMD,IGVFSM8043HCTC	094_CASTJ_10F_03	16	R	F7
CCTCATTA	IGVFSM7747QNMD,IGVFSM8043HCTC	094_CASTJ_10F_03	16	T	F7
TGGCTTCC	IGVFSM0611PVMT,IGVFSM8601VZGE	093_CASTJ_10M_03	16	R	F8
CTTATACC	IGVFSM0611PVMT,IGVFSM8601VZGE	093_CASTJ_10M_03	16	T	F8
TTGTTGCC	IGVFSM5009WOZO	040_129S1J_10F_06	16	R	F9
TCTATTAC	IGVFSM5009WOZO	040_129S1J_10F_06	16	T	F9
TTGTTGCC	IGVFSM4983NNIU	090_CASTJ_10F_06	16	R	F9
TCTATTAC	IGVFSM4983NNIU	090_CASTJ_10F_06	16	T	F9
GTCATCTC	IGVFSM5004FHRH	043_129S1J_10M_06	16	R	F10
CCTGCATT	IGVFSM5004FHRH	043_129S1J_10M_06	16	T	F10
GTCATCTC	IGVFSM2511KNVM	091_CASTJ_10M_06	16	R	F10
CCTGCATT	IGVFSM2511KNVM	091_CASTJ_10M_06	16	T	F10
TTGCTCAT	IGVFSM0279NVKF	044_129S1J_10F_06	16	R	F11
CAATCCTT	IGVFSM0279NVKF	044_129S1J_10F_06	16	T	F11
TTGCTCAT	IGVFSM8621PXJN	094_CASTJ_10F_06	16	R	F11
CAATCCTT	IGVFSM8621PXJN	094_CASTJ_10F_06	16	T	F11
CTGTCTGC	IGVFSM5577TDWZ	045_129S1J_10M_06	16	R	F12
TTGTCTTA	IGVFSM5577TDWZ	045_129S1J_10M_06	16	T	F12
CTGTCTGC	IGVFSM0998PUXL	093_CASTJ_10M_06	16	R	F12
TTGTCTTA	IGVFSM0998PUXL	093_CASTJ_10M_06	16	T	F12
TATATTCC	IGVFSM1395QLCT,IGVFSM2105LIWI	058_WSBJ_10F_03	16	R	G1
TCACTTTA	IGVFSM1395QLCT,IGVFSM2105LIWI	058_WSBJ_10F_03	16	T	G1
ATATTGGC	IGVFSM1203GHYB,IGVFSM6575IFHC	057_WSBJ_10M_03	16	R	G2
TGCTTGGG	IGVFSM1203GHYB,IGVFSM6575IFHC	057_WSBJ_10M_03	16	T	G2
GTGTCCTC	IGVFSM1996USDH,IGVFSM9916UWTV	060_WSBJ_10F_03	16	R	G3
CGCTCATT	IGVFSM1996USDH,IGVFSM9916UWTV	060_WSBJ_10F_03	16	T	G3
ATCTTCAT	IGVFSM1116FVJP,IGVFSM1295CUJX	059_WSBJ_10M_03	16	R	G4
GCCTCTAT	IGVFSM1116FVJP,IGVFSM1295CUJX	059_WSBJ_10M_03	16	T	G4
CGTGGTTG	IGVFSM2903TFAJ,IGVFSM5785PIAM	062_WSBJ_10F_03	16	R	G5
GAGCACAA	IGVFSM2903TFAJ,IGVFSM5785PIAM	062_WSBJ_10F_03	16	T	G5

TTGCATCC	IGVFSM0431DSEH,IGVFSM0898KHQX	061_WSBJ_10M_03	16	R	G6
CTCTTAAC	IGVFSM0431DSEH,IGVFSM0898KHQX	061_WSBJ_10M_03	16	T	G6
TCTTAATC	IGVFSM1901NWCD,IGVFSM7994PCAR	096_WSBJ_10F_03	16	R	G7
TCTAGGCT	IGVFSM1901NWCD,IGVFSM7994PCAR	096_WSBJ_10F_03	16	T	G7
TGCATTTC	IGVFSM1039KCGC,IGVFSM8727ZEXW	063_WSBJ_10M_03	16	R	G8
AATTCTGC	IGVFSM1039KCGC,IGVFSM8727ZEXW	063_WSBJ_10M_03	16	T	G8
GATGTTTC	IGVFSM0510YNAJ	046_NZOJ_10F_06	16	R	G9
CATTCTCA	IGVFSM0510YNAJ	046_NZOJ_10F_06	16	T	G9
GATGTTTC	IGVFSM3432ZCOX	058_WSBJ_10F_06	16	R	G9
CATTCTCA	IGVFSM3432ZCOX	058_WSBJ_10F_06	16	T	G9
ATCTTGTC	IGVFSM7577ZSZL	047_NZOJ_10M_06	16	R	G10
ACTTGCCT	IGVFSM7577ZSZL	047_NZOJ_10M_06	16	T	G10
ATCTTGTC	IGVFSM1456DXFQ	057_WSBJ_10M_06	16	R	G10
ACTTGCCT	IGVFSM1456DXFQ	057_WSBJ_10M_06	16	T	G10
TCATATTC	IGVFSM4204PXKJ	048_NZOJ_10F_06	16	R	G11
ATCATTGC	IGVFSM4204PXKJ	048_NZOJ_10F_06	16	T	G11
TCATATTC	IGVFSM7766OOSU	060_WSBJ_10F_06	16	R	G11
ATCATTGC	IGVFSM7766OOSU	060_WSBJ_10F_06	16	T	G11
TGGCCTCT	IGVFSM8056EVEE	049_NZOJ_10M_06	16	R	G12
GTTCAACA	IGVFSM8056EVEE	049_NZOJ_10M_06	16	T	G12
TGGCCTCT	IGVFSM6894ALPA	059_WSBJ_10M_06	16	R	G12
GTTCAACA	IGVFSM6894ALPA	059_WSBJ_10M_06	16	T	G12
CGTTGTCT	IGVFSM0322EOHO,IGVFSM4189AKUL	092_CASTJ_10F_03	16	R	H1
CCATTTGC	IGVFSM0322EOHO,IGVFSM4189AKUL	092_CASTJ_10F_03	16	T	H1
TCTTGTCA	IGVFSM7371QPPK,IGVFSM9072COND	047_NZOJ_10M_03	16	R	H2
GACTTTGC	IGVFSM7371QPPK,IGVFSM9072COND	047_NZOJ_10M_03	16	T	H2
TATTCCTG	IGVFSM8490EBUM,IGVFSM9852CDAM	048_NZOJ_10F_03	16	R	H3
ATTGGCTC	IGVFSM8490EBUM,IGVFSM9852CDAM	048_NZOJ_10F_03	16	T	H3
TCCATGTC	IGVFSM0101ITFE,IGVFSM4528ONTR	049_NZOJ_10M_03	16	R	H4
GTGCTAGC	IGVFSM0101ITFE,IGVFSM4528ONTR	049_NZOJ_10M_03	16	T	H4
TTGTCATC	IGVFSM0561GQRI,IGVFSM5306QAYC	050_NZOJ_10F_03	16	R	H5
CTTTCAAC	IGVFSM0561GQRI,IGVFSM5306QAYC	050_NZOJ_10F_03	16	T	H5
ATTTCCTG	IGVFSM1083MRIY,IGVFSM5958RVMJ	051_NZOJ_10M_03	16	R	H6
ACTATTGC	IGVFSM1083MRIY,IGVFSM5958RVMJ	051_NZOJ_10M_03	16	T	H6
GTGTCTCC	IGVFSM6096DCNO,IGVFSM8959FRNP	052_NZOJ_10F_03	16	R	H7

ACTGGCTT	IGVFSM6096DCNO,IGVFSM8959FRNP	052_NZOJ_10F_03	16	T	H7
GTGTGTGT	IGVFSM8856ZHGA,IGVFSM9234QGBN	053_NZOJ_10M_03	16	R	H8
ATTAGGCT	IGVFSM8856ZHGA,IGVFSM9234QGBN	053_NZOJ_10M_03	16	T	H8
TATGCTTC	IGVFSM8878JBBO	050_NZOJ_10F_06	16	R	H9
GCCTTTCA	IGVFSM8878JBBO	050_NZOJ_10F_06	16	T	H9
TATGCTTC	IGVFSM3113AHHA	062_WSBJ_10F_06	16	R	H9
GCCTTTCA	IGVFSM3113AHHA	062_WSBJ_10F_06	16	T	H9
ATGGTGTT	IGVFSM2214BDBR	051_NZOJ_10M_06	16	R	H10
ATTCTAGG	IGVFSM2214BDBR	051_NZOJ_10M_06	16	T	H10
ATGGTGTT	IGVFSM8737KUQW	061_WSBJ_10M_06	16	R	H10
ATTCTAGG	IGVFSM8737KUQW	061_WSBJ_10M_06	16	T	H10
GAATAATG	IGVFSM8146OBNU	052_NZOJ_10F_06	16	R	H11
CCTTACAT	IGVFSM8146OBNU	052_NZOJ_10F_06	16	T	H11
GAATAATG	IGVFSM6802LFPK	096_WSBJ_10F_06	16	R	H11
CCTTACAT	IGVFSM6802LFPK	096_WSBJ_10F_06	16	T	H11
CCTCTGTG	IGVFSM4952NHSA	053_NZOJ_10M_06	16	R	H12
ACATTTGG	IGVFSM4952NHSA	053_NZOJ_10M_06	16	T	H12
CCTCTGTG	IGVFSM6965RRET	063_WSBJ_10M_06	16	R	H12
ACATTTGG	IGVFSM6965RRET	063_WSBJ_10M_06	16	T	H12

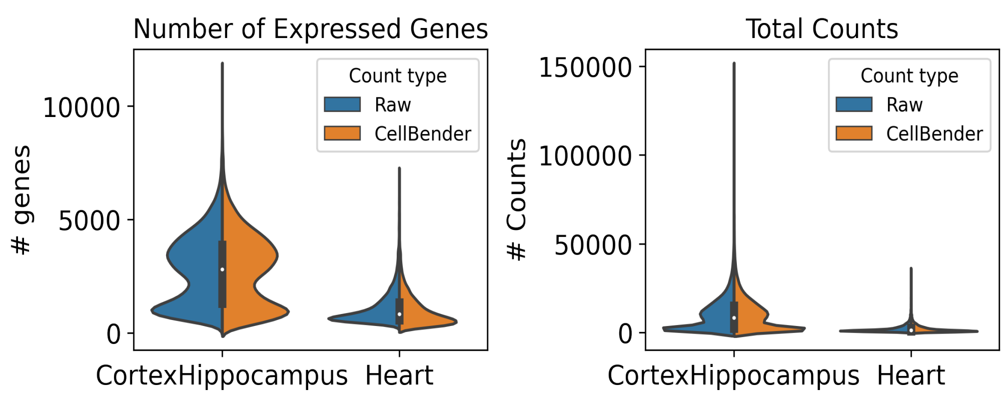
6. Description of data processing workflow and h5ad files

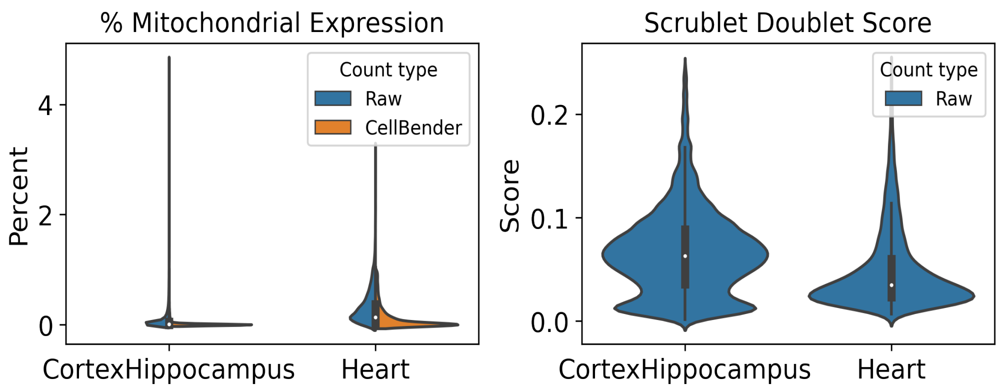
Data from Subpool_9 were combined with all other subpools into one AnnData object per tissue as well as matching data from other plates for processing and cell type annotation. The AnnData contains detailed sample and cell metadata in the observation (obs) table and gene information in the variables table (var). Cell IDs (adata.obs.index) are re-formatted to be human-readable and unique across subpools and experiments by appending subpool and experiment IDs. The final AnnData contains both raw and CellBender denoised counts matrices in separate layers: adata.layers['raw_counts'] and adata.layers['cellbender_counts']. The current adata.X points to the CellBender denoised counts. Cells or nuclei are filtered by >0.5 cell_probability from CellBender analysis.

In addition to CellBender filtering, nuclei were also filtered by the following QC parameters:

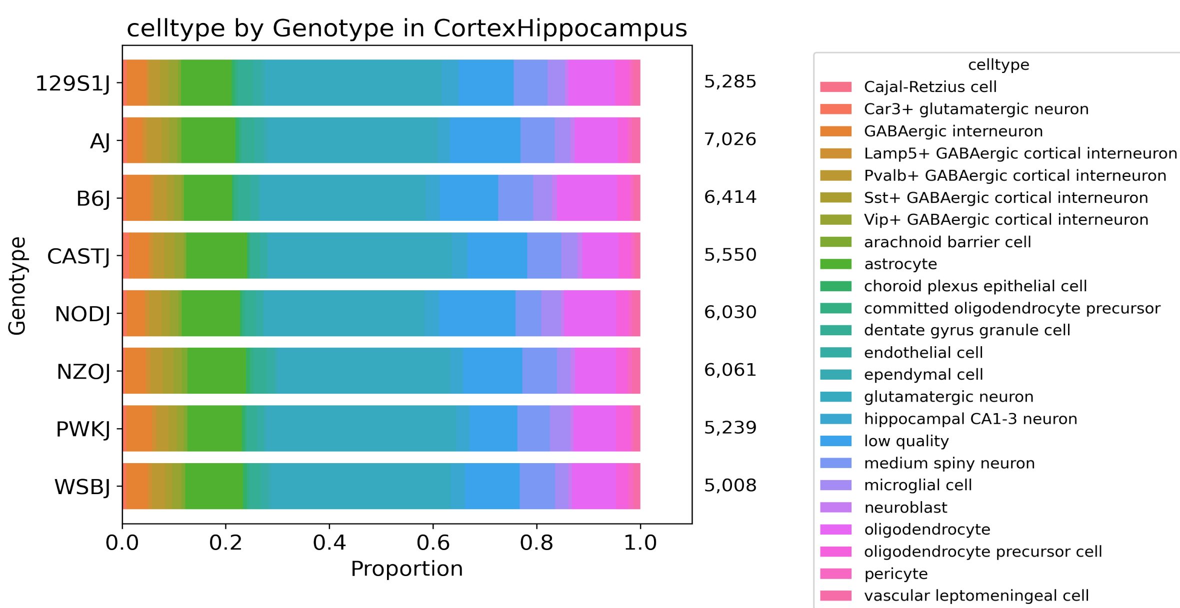
Table 8: QC thresholds

Parameter	Threshold
Minimum # UMIs	500.0
Maximum # UMIs	150000.0
Minimum # genes	250.0
Maximum % mito.	1.0
Maximum doublet score	0.25





After filtering, data for each tissue was normalized and clustered using CellBender denoised counts. Normalization includes regression of technical parameters (number of genes expressed per cell and percent mitochondrial gene expression). Harmony was used to integrate exome capture and non-exome capture subpools for annotation. A Leiden clustering resolution of 1 was used for 55 total clusters in CortexHippocampus and 33 clusters in Heart, using all tissue data across experiments. Cell type annotations were assigned per Leiden cluster or subcluster (leiden_R) using expression of canonical cell type marker genes.



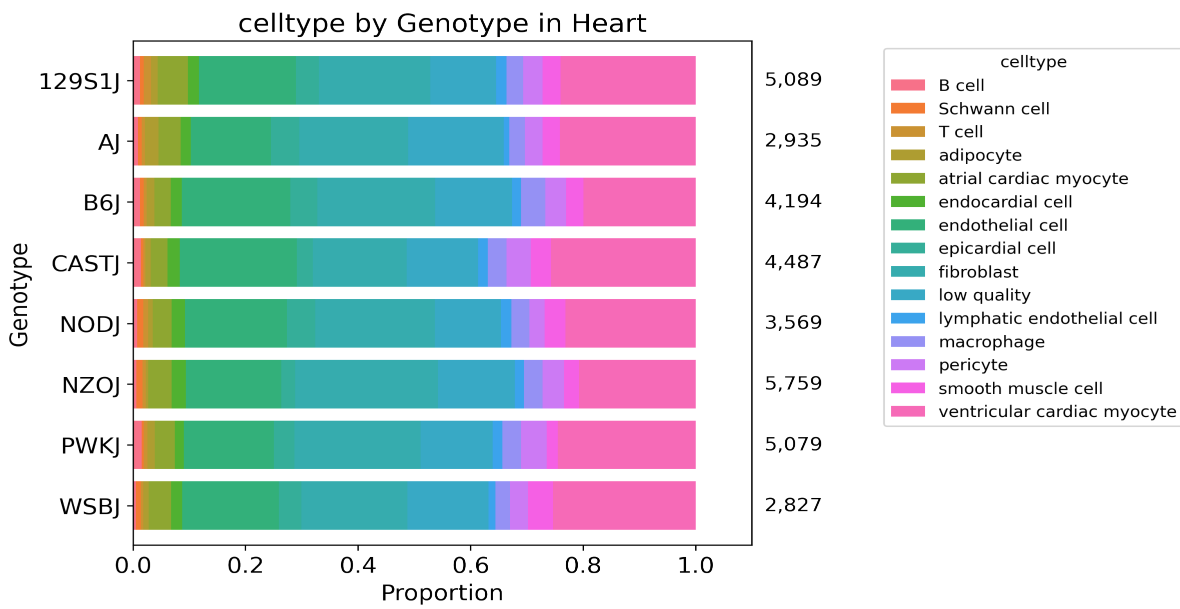


Table 9: Description of obs keys in h5ad file

Key	Description
cellID	Index of obs table. Includes barcoding wells for each round (1_2_3), subpool, and experiment.
lab_sample_id	Human-readable sample name used in-house.
sample	IGVF tissue accession.
plate	Experiment ID, e.g. igvf_003.
subpool	Subpool ID, e.g. Subpool_2. Number corresponds to Illumina index ID from Parse Bio kit.
SampleType	Nuclei or Cells.
Tissue	Human-readable tissue name, e.g. Kidney.
Sex	Male or Female.
Age	Age in postnatal months (PNM) of the mouse, e.g. PNM_02.
Genotype	Mouse genotype/strain, e.g. B6J.
leiden	Numeric Leiden cluster starting from 0.
leiden_R	If necessary, Leiden clusters and any sub-clusters, e.g. 15_1.
subpool_type	Exome capture (EX) or no exome capture (NO).
general_celltype	Broadest level of cell type annotations, e.g. neuron.
general_CL_ID	EMBL CL database ID for general cell type annotation, e.g. CL:0000540.
celltype	Finest level of cell type annotation with a CL ID, e.g. Vip+ GABAergic cortical interneuron.
CL_ID	EMBL CL database ID for cell type annotation, CL:4023016
subtype	Finest level of cell type annotation that may not have a CL ID. Mostly matches celltype.
Protocol	Cell barcoding and library building protocol/kit, e.g. Parse_WT.

Chemistry	Protocol chemistry version, v2 or v3.
bc	24-nucleotide sequence corresponding to 3 8-nucleotide barcode rounds (3, 2, 1).
bc1_sequence	8-nucleotide sequence barcode for round 1.
bc2_sequence	8-nucleotide sequence barcode for round 2.
bc3_sequence	8-nucleotide sequence barcode for round 3.
bc1_well	Round 1 well.
bc2_well	Round 2 well.
bc3_well	Round 2 well.
Row	Sample barcoding plate (round 1) row.
Column	Sample barcoding plate (round 1) column.
well_type	Single or Multiplexed.
Mouse_Tissue_ID	Same as lab_sample_id.
Multiplexed_sample1	Sample 1 from multiplexed well (if applicable)
Multiplexed_sample2	Sample 2 from multiplexed well (if applicable)
DOB	Date of birth of the mouse in month/day/year format.
Age_days	Age of the mouse in days.
Body_weight_g	Body weight in grams of the mouse.
Estrus_cycle	Estimated estrus stage of the mouse, if applicable.
Dissection_date	Date the tissue was harvested in month/day/year format.
Dissection_time	Time in hours:minutes (24-hour time) the mouse was sacrificed.
ZT	Number of hours into the light cycle, from lights on (06:30, ZT0) to lights off (20:30, ZT14).
Dissector	Initials of technician who dissected the tissue.
Tissue_weight_mg	Tissue weight in milligrams.
Notes	Notes taken during sample collection, experiment, or data processing.
n_genes_by_counts_raw	The number of genes with at least 1 count, calculated using raw counts.
total_counts_raw	Total counts (UMIs), calculated using raw counts.
total_counts_mt_raw	Total counts for mitochondrial genes, calculated using raw counts.
pct_counts_mt_raw	Percent of total counts in mitochondrial genes, calculated using raw counts.
n_genes_by_counts_cb	The number of genes with at least 1 count, calculated using CellBender counts.
total_counts_cb	Total counts (UMIs), calculated using CellBender counts.
total_counts_mt_cb	Total counts for mitochondrial genes, calculated using CellBender counts.
pct_counts_mt_cb	Percent of total counts in mitochondrial genes, calculated using CellBender counts.
doublet_score	Scrublet doublet score, calculated using CellBender counts.
predicted_doublet	Scrublet predicted doublet (True or False).
background_fraction	Fraction of counts CellBender predicted to be background noise.

cell_probability	CellBender probability that the cell is real.
cell_size	CellBender size factor.
droplet_efficiency	CellBender statistic indicating transcript capture efficiency per cell.
B6J_klue_counts	Estimated counts for B6J genotype from klue analysis.
NODJ_klue_counts	Estimated counts for NODJ genotype from klue analysis.
AJ_klue_counts	Estimated counts for AJ genotype from klue analysis.
PWKJ_klue_counts	Estimated counts for PWKJ genotype from klue analysis.
129S1J_klue_counts	Estimated counts for 129S1J genotype from klue analysis.
CASTJ_klue_counts	Estimated counts for CASTJ genotype from klue analysis.
WSBJ_klue_counts	Estimated counts for WSBJ genotype from klue analysis.
NZOJ_klue_counts	Estimated counts for NZOJ genotype from klue analysis.

7. Package versions

Table 10: Package versions

package	version
kallisto	0.50.1
bustools	0.43.2
kb	0.28.2
Python	3.9.0
snakemake	7.32.0
cellbender	0.3.2
scanpy	1.10.2
scrublet	0.2.3
anndata	0.10.8
pandas	2.2.2
numpy	1.26.4
harmonypy	0.0.9

8. Contact information and GitHub

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Pipeline repo: https://github.com/mortazavilab/parse_pipeline

Code used to make this report: https://github.com/mortazavilab/8cube_manuscript/blob/main/plate_report/Subpool_report_igvf_003.ipynb

Code used to process and annotate the data for main tissue: https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/CortexHippocampus.ipynb

Code used to process and annotate the data for multiplexed tissue: https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/Heart.ipynb